

ECONOMICS & FINANCE

Time Series Basics in 6 Hours

A presentation version for economics and finance undergraduates

Time order · Level vs change · Stationarity · Autocorrelation · Forecast evaluation

The goal today is not to memorise model names. It is to internalise the order in which an economic or financial time series should be examined.

Student handout edition · five practice datasets included

The Opening Question

A research intern's first day

“We have KOSPI, USD/KRW and CPI data. Can any of it help us explain tomorrow, or next month?”

Four questions to ask before running a model

- ① Is there an order to the data?
- ② Are we looking at a level or a change?
- ③ Is the pattern stable?
- ④ Does yesterday explain today?

The Workflow We Repeat for Six Hours

Learn the order once, then apply it to many series



The point is to compare fairly, respecting time order, before claiming that anything was predicted.

The Five Sentences That Should Remain

We return to these five sentences all day

- ① Time order is information.
- ② A level and a change are different things.
- ③ In finance, look at returns first.
- ④ Check whether yesterday explains today.
- ⑤ Evaluate a forecast on future data, against a benchmark.

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HOUR 1 • TIME ORDER

A Time Series Is an Ordered Story

Shuffle the scenes and the plot disappears

The Order Is the Information

Some questions become unanswerable the moment rows are shuffled

CPI runs January, then February, then March. Shuffle those rows and the question “how did prices change?” can no longer be answered.

Safe to shuffle

- Height and weight of 100 students
- A list of survey respondents
- Rows that stand on their own

Never shuffle

- Monthly CPI
- Daily index and exchange rates
- Quarterly GDP

Ordinary Data versus Time-Series Data

The table looks the same; the question is not

	Ordinary data	Time-series data
Example	Height and weight of 100 students	Monthly CPI, daily index, daily FX
Row order	Meaning survives a reshuffle	Trend and change disappear
Central question	Who is larger?	How did it change from before?

When a time series arrives, the first action is not estimation. It is drawing the graph.

Practice 1: Plot First

See the story before calculating anything

Input files

- kospi_demo.csv
- usdkrw_demo.csv
- cpi_demo.csv

Tasks

- Convert Date to a datetime
- Plot the level series
- Describe the movement in words

Questions

- Does it keep rising?
- Does it swing up and down?
- Are there odd values?

```
df['Date'] = pd.to_datetime(df['Date'])  
df.set_index('Date')['Close'].plot()
```

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HOUR 2 · LEVEL VS CHANGE

A Level and a Change Are Different

KOSPI 2,500 alone does not tell us whether today was good

What Does KOSPI 2,500 Mean?

Not much, until we know yesterday's value

If yesterday was 2,450 the market rose. If yesterday was 2,550 it fell. The same number, two opposite days.

Where we are · level

- KOSPI 2,500
- USD/KRW 1,350
- CPI 114.2

How we moved · change

- Up 2% or down 2% from yesterday?
- How much above last month?
- How much above last year?

Level Questions and Change Questions

The same series answers two very different questions

Data	Level question	Change question
Equity index	How many points is KOSPI?	How many percent did it move today?
Exchange rate	What is USD/KRW?	How much did the won move?
CPI	What is the price index?	How much did prices rise from last year?
GDP	How large is the economy?	How fast did the economy grow?

How to Compute a Change and a Return

A feel for the numbers comes before the formula

$$\text{change} = \text{today's value} - \text{yesterday's value}$$

$$\text{return} = \text{today's price} \div \text{yesterday's price} - 1$$

A market example

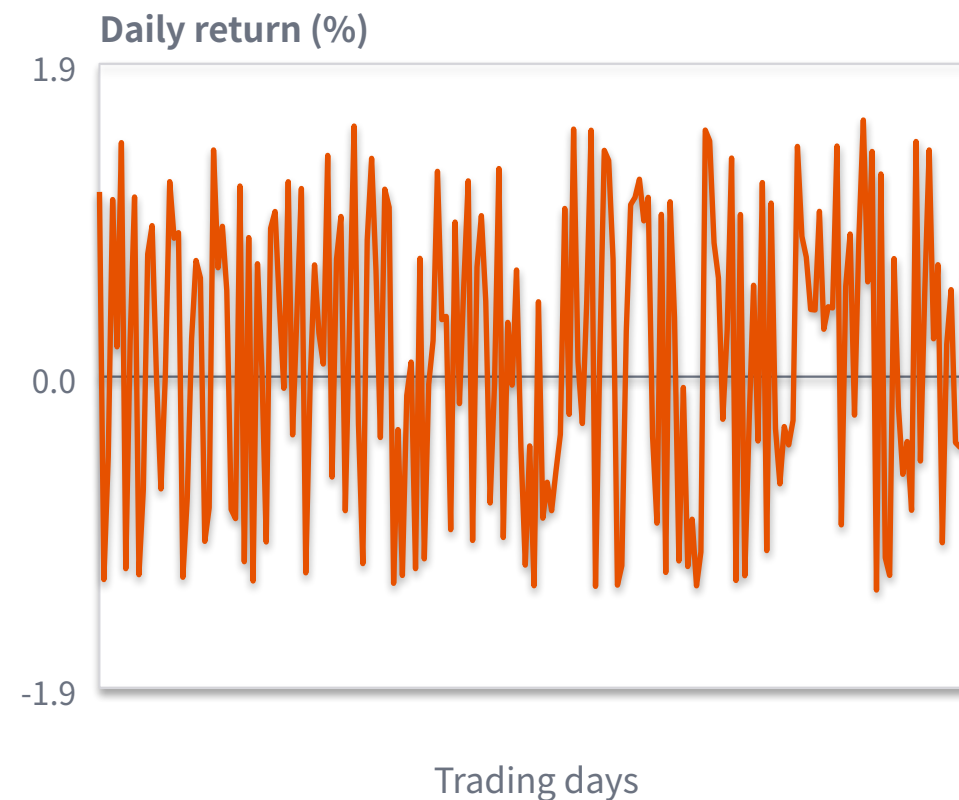
- A price moves from 100 to 102.
- The change is +2; the return is +2%.

An economics example

- CPI moves from 110 to 112.2.
- Prices rose by about 2%.

Price Level and Return Tell Different Stories

The same data, plotted two ways



The left panel shows a long-run path; the right panel moves around zero.

Practice 2: Turn a Level into a Change

The single most important habit in financial analysis

```
df['Change'] = df['Close'].diff()  
df['Return'] = df['Close'].pct_change()
```

A sample sentence to submit

- The price level shows a long-run path, while the return moves around 0%.
- For short-horizon work, therefore, we analyse the return series first.

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HOUR 3 · STATIONARITY

Does a Similar Environment Repeat?

A moving escalator versus a flat corridor

The Escalator and the Corridor

The same ball behaves differently depending on the floor

A moving escalator

- The escalator's rise mixes into the motion.
- The average position keeps shifting.
- Past averages are hard to trust.

A flat corridor

- The ball moves around a centre.
- The average position stays put.
- Past averages and swings are usable.

Stationarity asks which of these two the series in front of you resembles.

Stationarity in Plain English

Not perfect stability — just stability we can work with

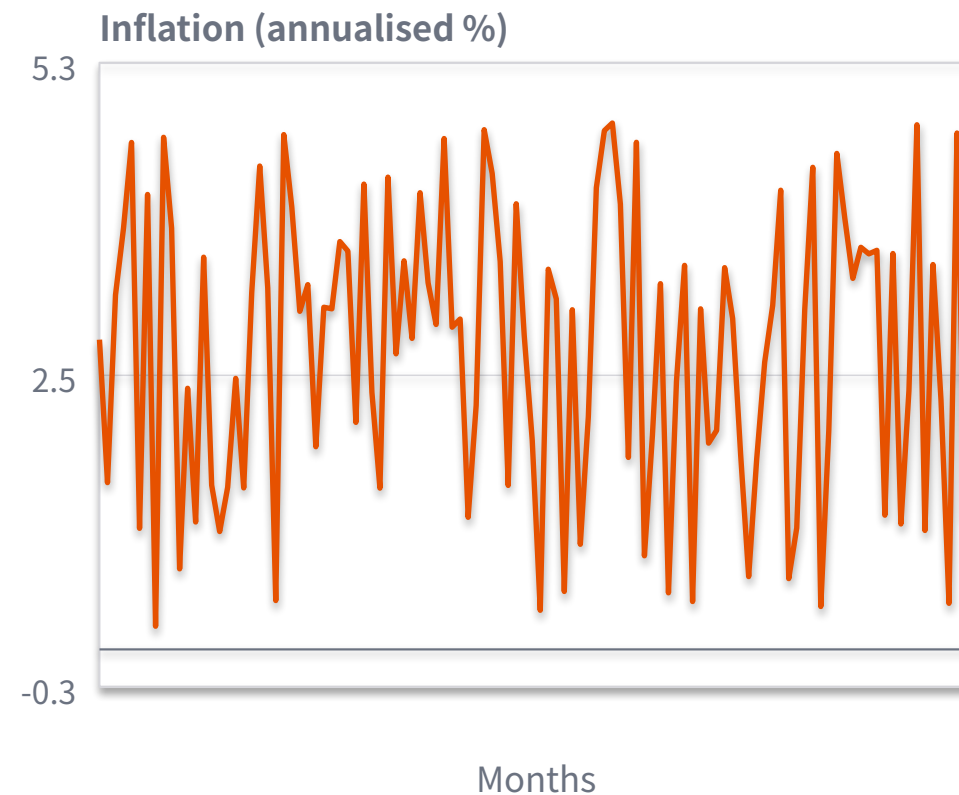
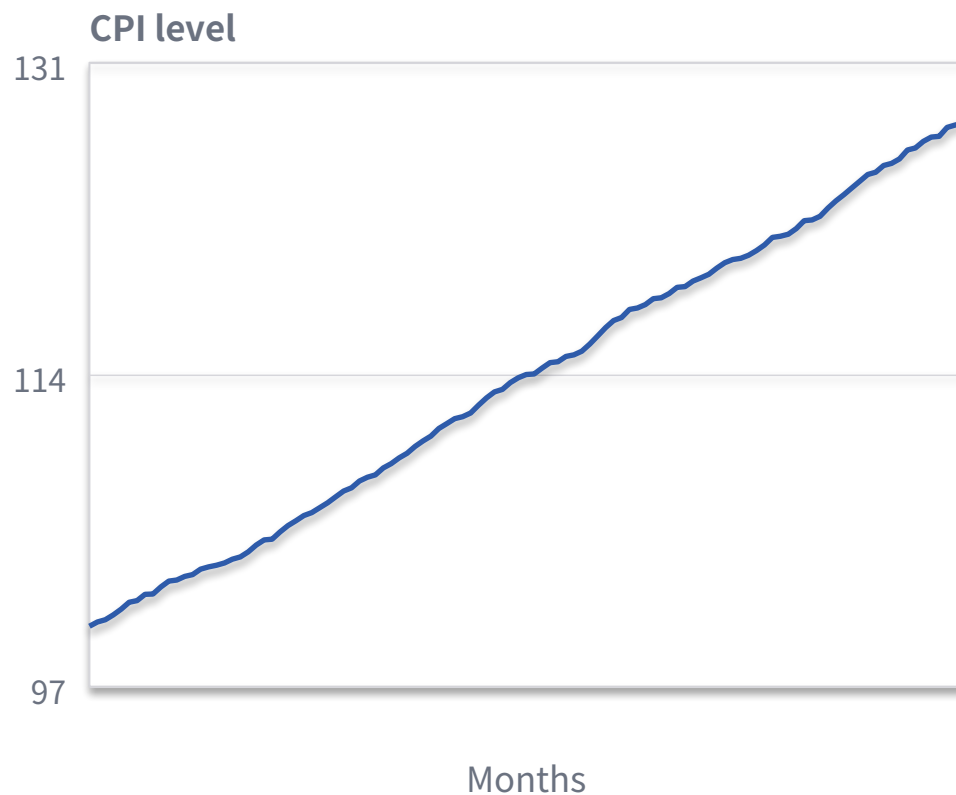
The average position and the typical size of the fluctuations do not change much over time.

Why it matters

- It decides whether past averages and swings can inform a judgement about the future.
- If the environment keeps shifting, the past average is background information at best.
- So we check it before choosing a model.

CPI Level and the Inflation Rate

The same distinction applies to economic data



The level keeps climbing; the rate of change moves inside a band.

Why We Look at Returns First

They are usually easier to work with than price levels

If we keep the level

- The average keeps moving.
- No past average serves as a reference.

If we take the change

- The series moves around zero.
- Past averages and swings become comparable.

Easier to work with, however, does not mean accurate. That is why forecast evaluation exists.

Practice 3: Judge Stability by Eye First

Put the level plot and the change plot side by side

Tasks

- ① Plot the price level.
- ② Plot the return series.
- ③ Say which one moves around an average.

Sample interpretation

- The price level shows a long-run path.
- The return moves around 0%.
- So the next step uses the return series.

Keep the code short and the interpretation explicit.

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HOUR 4 · AUTOCORRELATION

Does Yesterday Connect to Today?

A market offers three possibilities

Three Possibilities

Autocorrelation is how we check, with a number, which one is closest

Momentum

- Yesterday went up,
- and today may go up too.

Reversal

- Yesterday rose sharply,
- so today may give some back.

Weak link

- Yesterday's information
- barely explains today.

Financial returns are usually closest to the third case. That is exactly why forecasts must be tested.

Lag 1 Is the Previous Observation

Yesterday in daily data; last month in monthly data



To explain today, t , we use the value immediately before it, $t-1$.

```
df['Return_lag1'] = df['Return'].shift(1)
```

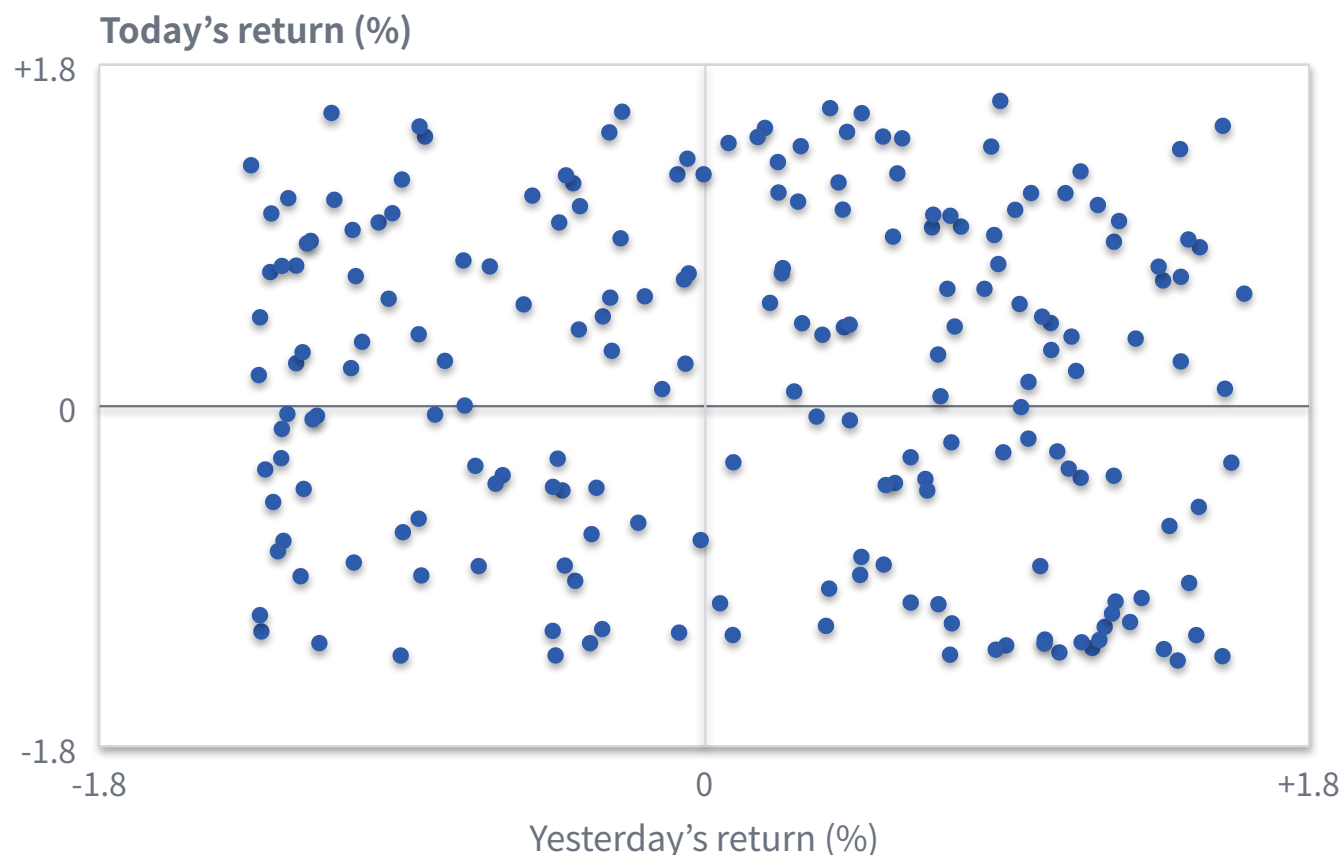
One Correlation Gives the Direction

Compare yesterday's return with today's

Correlation	Plain reading	Sample sentence
Positive	Yesterday and today tend to move the same way	Days after a rise are a little more often up
Negative	Yesterday and today tend to move opposite ways	A rise is sometimes partly given back the next day
Near zero	Yesterday alone explains little about today	Yesterday's return has weak predictive power

Yesterday's Return versus Today's

Do the points form a clear line?



What to look for

- Do points bunch toward the upper right?
- Do they slope from upper left to lower right?
- Or are they simply scattered?

A cloud of points means yesterday alone explains little about today.

A Very Simple Model

If you have seen regression, you already understand this

$$\text{today's return} = a + b \times \text{yesterday's return} + \text{error}$$

What b looks like

Reading

b is positive

Yesterday's move may continue today

b is negative

Part of yesterday's move may be given back

b is near zero

Yesterday's return explains little

Practice 4: Does Yesterday Explain Today?

Check the correlation and the simple regression together

```
X = sm.add_constant(df['Return_lag1'])  
model = sm.OLS(df['Return'], X, missing='drop').fit()
```

What to interpret

- ① Is b positive or negative?
- ② Is it large or small?
- ③ Does it beat the benchmark?

What to remember

- The output table is not the point.
- The question is whether past information genuinely helps.

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HOUR 5 · EVALUATION

A Forecast Must Beat a Benchmark

An honest comparison matters more than an impressive model

Fix the Benchmark First

Without something to compare against, “good” has no meaning

For returns, the benchmark “tomorrow’s return is 0%” is stronger than most people expect.

Target

The simplest benchmark

Price

tomorrow’s price = today’s price

Return

tomorrow’s return = 0%

CPI inflation

next month = the recent average

Respect Time Order When Evaluating

A random split means peeking at the future

TRAIN · earlier 80%

TEST · last 20%

time →

The rule

- Train on the earlier stretch and keep the later stretch aside as if it were the future.

Common mistakes

- Shuffling rows before splitting.
- Standardising over the whole sample first.
- Both leak the future into training.

Smaller Errors Are Better

MAE and RMSE tell us how wrong we were

MAE

- The average absolute error
- How wrong are we on a typical day?

RMSE

- The root of the mean squared error
- Large misses weigh much more heavily

Report both, and always write them next to the benchmark's values.

Practice 5: Model versus Benchmark

State “beats it” or “does not” with numbers

Tasks

- ① Train on the first 80%, test on the last 20%.
- ② Predict today from yesterday's return.
- ③ Let the benchmark predict 0%.
- ④ Compare MAE and RMSE.

Sample conclusion

- In this sample the simple model beat the benchmark.
- The margin, however, was small.

If the margin is small, write that it is small. That is a result too.

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HOUR 6 • MINI PROJECT

Run the Whole Workflow on One Dataset

Walk the path you learned, from start to finish

The Mini Project in Five Steps

Leave one written sentence at every step

Step	Task	Sample sentence
1	Plot the level	The price level shows a long-run path
2	Build the change series	The return moves around 0%
3	Build lag 1	Compare yesterday's return with today's
4	Evaluate the forecast	Compare the errors with the benchmark
5	Write the conclusion	Past information alone gives weak prediction

What to Submit

The interpretation matters more than the code

- ① One level plot
- ② One change plot
- ③ A lag-1 scatter or correlation
- ④ An error table against the benchmark
- ⑤ Three sentences of conclusion

Rules for Students

Read this slide before starting the practice

- ① Plot before modelling, and describe what you see.
- ② Always separate level from change.
- ③ Never shuffle time order.
- ④ Compare every model with a benchmark.
- ⑤ Write interpretation, not just code, in what you hand in.

Final Recap

If these five sentences remain, the course succeeded

- ① Time order is information.
- ② A level and a change are different things.
- ③ In finance, look at returns first.
- ④ Check whether yesterday explains today.
- ⑤ Evaluate a forecast on future data, against a benchmark.

ONE SENTENCE

Time order is information.

Plot before modelling, separate level from change, and compare against a benchmark. Following that order alone prevents most of the wrong conclusions.